

You need to create a minimal but complete web_application. You can use any language, framework, database to complete this task.

You have to think about all the apis necessary and the database design for the problem. We want the front end to be in React. They both have to be integrated with each other.

You will be evaluated on correctness of code, database design, api design, and how clean and modular code is.

Any design_documents and api_documentations will be appreciated.

We want to see a Demo – feel free to create a demo video.

We will test the app using Postman Api and the front end.

The task is to design a project management application like Trello.

The app has 4 main components Users, Boards, BoardLists, Cards

Users can be added to the application

Each User should have these attributes [id, name, email]

Board signifies a project

Boards can be created and added to the application

Users can be added to a Board

Each Board should have these attributes [id, name, privacy("PUBLIC", "PRIVATE"), url]

A Board can have many BoardLists

BoardList signifies a sub_project

BoardLists can be created and added to a Board

Cards signifies individual tasks

Cards can be created and added to a BoardList

A Card can be assigned a User who is present in the Board

Each Card should have these attributes [id, name, description]

Features

Board Features

1. We should be able to create / delete boards.

2. Add users to a board.
3. Update board attributes.
4. Deleting a board should delete all board_lists inside it.

BoardList Features

1. We should be able to create / delete board_lists.
2. Update board_list attributes.
3. Deleting a board_list should delete all cards inside it.

Card Features

1. We should be able to create / delete cards.
2. Update card attributes.
3. Assign and unassign a user to card.

Other Features

1. We should be able to move cards across lists in the same board
2. Need to have apis to get all boards, get one board, get one board_list, and get one card
3. Default board privacy should be public
4. Cards should be unassigned by default
5. Urls should be based on Id